

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Computer Science & Engineering

Information System Design Laboratory

Industrial Visit Report

Recognition of Need

Industry Visited: Brain Station 23

Submitted To

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1. Introduction

1.1 Background of the Industrial Visit

As part of the Information System Design Laboratory course, students of the Department of Computer Science and Engineering undertook a structured industrial visit to Brain Station 23, a leading software development company headquartered in Dhaka, Bangladesh. This visit was organized with the academic objective of exposing students to real-world practices of information system design, software engineering, and organizational management within a professional software firm. The visit provided a rare opportunity to bridge the gap between theoretical classroom instruction and the practical demands of a modern IT organization.

1.2 Overview of Brain Station 23

Brain Station 23 is a premier software engineering organization with over 20 years of experience in the global software industry. With a workforce exceeding 1,000 employees, the organization operates direct branches in six countries: Bangladesh, the United States of America, the United Arab Emirates, Malaysia, Germany, and Japan. The company has established itself as a significant player in enterprise software development, particularly in the domain of Enterprise Resource Planning (ERP) systems. Its clientele includes globally recognized corporations such as Toyota, Nissan, PayPal, MetLife, and Unipaid. Brain Station 23 is ISO 9000 and ISO 27000 certified and holds CMMI Level 3 accreditation, reflecting its commitment to quality and process maturity. Furthermore, it is an AWS Advanced Consulting Partner and has received the Google AI Competition Bronze Award on two separate occasions.

1.3 Purpose of the Visit

The primary purpose of the visit was to observe and analyze how Brain Station 23 identifies, documents, and responds to organizational and client-driven needs through structured information systems. The visit also aimed to understand the company's adoption of globally recognized software development methodologies, including the Agile framework and SCRUM practices, and to observe how these methodologies are applied to manage complex, large-scale projects across multiple international markets.

1.4 Importance of Recognition of Need in Information System Design

Recognition of Need is the foundational phase in the Information System Development Life Cycle (ISDLC). Before any system can be designed or developed, there must be a clear, analytically grounded understanding of the problems, inefficiencies, or gaps that the system is intended to resolve. Without this critical first step, development efforts risk misalignment with business objectives, resulting in costly failures. The industrial visit to Brain Station 23 offered concrete, real-world examples of how a large-scale software organization systematically recognizes, documents, and acts upon these needs at both strategic and operational levels.



2. Problem

Through information gathered during the interactive Q&A session with Brain Station 23 representatives, as well as logical inferences drawn from the operational context of the organization, the following critical problems were identified. Each problem is analyzed in terms of its nature, its implications for information system design, and its relevance to software development practices at Brain Station 23.

2.1 Requirement Management and Change Control

Problem Description

During the Q&A session, it was disclosed that Brain Station 23 follows a formal process whereby requirement changes are placed into a structured queue rather than being implemented ad hoc. The Business Analyst (BA), who also performs the role of Scrum Master, is responsible for documenting requirements through a Software Requirement Specification (SRS) document. However, in a dynamic business environment serving multinational clients such as Toyota and Nissan, requirements are subject to frequent revision due to evolving business strategies, regulatory changes, and shifting user expectations.

Need for an Information System

Uncontrolled requirement changes, commonly known as scope creep, are among the most damaging challenges in software development. A robust requirement management system — anchored by the SRS — is essential to ensure that all

stakeholders maintain a consistent, version-controlled understanding of project scope. The formalization of change requests into a queue at Brain Station 23 represents a deliberate information system design decision that ensures traceability, accountability, and controlled evolution of software products.

2.2 Communication Gaps in Multi-Stakeholder Workflows

Problem Description

Brain Station 23 follows a specific workflow for project communication: the client first engages the Sales Team, which then liaises with the Business Analyst, who subsequently coordinates with the Technical Team. Following development milestones, the output is reviewed by the customer. While this structured chain of communication minimizes misinterpretation, it inherently introduces latency in information transfer. The risk of communication gaps increases substantially when projects involve teams distributed across Bangladesh, USA, UAE, Germany, Malaysia, and Japan.

An additional question posed during the visit — "How are miscommunications between the business analyst and the technical team resolved?" — revealed that Brain Station 23 employs an internal support and communication platform referred to as the Excellent Support System (ESS), which helps document and track inter-team communications and issue resolutions.

Need for an Information System

A centralized communication and issue-tracking information system such as ESS is indispensable in multi-stakeholder environments. Without such a system, critical project information may be lost, misinterpreted, or duplicated, leading to rework, missed deadlines, and client dissatisfaction. The need for an integrated platform that consolidates communication, task assignment, and documentation is a direct recognition of the organizational need arising from multi-layered, geographically distributed operations.

2.3 Project Scalability and Resource Allocation

Problem Description

With a workforce of over 1,000 employees and simultaneous projects for diverse global clients — including the development of the Grameenphone App, Banglalink App, Shwapno E-commerce Platform, Nissan AR Chatbot, and various VR-based applications — Brain Station 23 faces substantial challenges in dynamically allocating human and technical resources. During the Q&A session, the organization acknowledged that different projects demand varying skill sets, and that matching available talent to project requirements in real time is a complex operational challenge.

Need for an Information System

Effective resource allocation in a large organization requires a well-designed human resource management information system (HRMIS) integrated with project management tools. Such a system must maintain an updated repository of employee competencies, current workload, and availability, enabling managers to make data-driven allocation decisions. The recognition of this need at Brain Station 23 directly justifies investment in integrated enterprise-level project and resource management platforms.

2.4 Quality Assurance and Compliance

Problem Description

Brain Station 23's adherence to ISO 9000 (Quality Management Systems) and ISO 27000 (Information Security Management Systems) standards, combined with its CMMI Level 3 certification, reflects a deep institutional commitment to quality and security. However, maintaining compliance across multiple projects, development teams, and international branches is inherently demanding. The organization must continuously document processes, conduct internal audits, track defects, and ensure that security protocols are uniformly applied — a challenge that becomes exponentially complex at scale.

Need for an Information System

Quality assurance without information system support is neither scalable nor reliable. Automated testing frameworks, continuous integration and deployment (CI/CD) pipelines, defect tracking systems, and audit management tools are all manifestations of recognized organizational needs for QA information systems. The ISO certifications held by Brain Station 23 are not merely symbolic; they mandate the existence and consistent use of documented information systems for quality management.

2.5 Customer Change Requests and Iterative Delivery

Problem Description

The company follows a 15-day client update cycle, providing progress reports and receiving feedback at regular intervals. This iterative model, aligned with Agile philosophy, means that customer-driven changes are frequent and expected. However, managing and prioritizing a continuous stream of change requests — particularly from high-profile clients such as MetLife or PayPal — without a formal information system creates the risk of version conflicts, feature regression, and inconsistent product delivery.

Need for an Information System

A change request management system, integrated with sprint planning tools within the Agile-SCRUM framework, is essential to systematically capture, evaluate, prioritize, and implement client-requested modifications. At Brain Station 23, this need is recognized through the formalized queue-based change management process and the 15-day review cycle, both of which are operationalized through dedicated information systems.

2.6 Global Team Coordination and Time Zone Management

Problem Description

Operating branches across six countries spanning multiple time zones introduces significant logistical complexity. A question raised during the visit — "How does Brain Station 23 coordinate development activities across teams in different time zones?" — prompted a discussion around the organization's use of asynchronous communication tools and structured sprint review schedules. Despite these measures, the potential for coordination failures — such as delayed code reviews, conflicting commits, or missed stand-up meetings — remains a legitimate operational concern.

Need for an Information System

Global software organizations require version control systems (such as Git), cloud-based project management platforms, asynchronous documentation systems, and automated notification frameworks to function cohesively across time zones. The recognition of this need has driven Brain Station 23 to leverage AWS cloud infrastructure and globally accessible collaboration tools, underscoring the indispensability of information systems in international software operations.

2.7 Information Security and Data Governance

Problem Description

As an ISO 27000-certified organization, Brain Station 23 is obligated to maintain rigorous information security standards across all operations. Given that the company handles sensitive financial data for clients such as PayPal and MetLife, and transactional data for platforms like Unipaid, data breaches, unauthorized access, and non-compliance with international data protection regulations represent existential operational risks.

Need for an Information System

Information security is not merely a technical concern but a systemic organizational need that demands dedicated information systems for access control, encryption, audit logging, and incident response. The ISO 27000 certification at Brain Station 23 institutionally recognizes this need and mandates the deployment of a comprehensive Information Security Management System (ISMS) as a non-negotiable operational requirement.

3. Necessity

The problems identified in the preceding section collectively establish a compelling case for the necessity of well-designed information systems at Brain Station 23. This section articulates why each identified problem necessitates a systematic, technology-driven solution.

3.1 Necessity of the Software Requirement Specification (SRS)

The SRS serves as the contractual and technical backbone of any software development project. At Brain Station 23, the necessity of the SRS arises from the need to establish an unambiguous, agreed-upon definition of system functionality before development begins. Given the complexity of projects such as the Nissan AR Chatbot or the Shwapno E-commerce Platform, a comprehensive SRS prevents misalignment between client expectations and delivered output. It also provides a legal reference point in the event of scope disputes, making it an indispensable information artifact in the development lifecycle.

3.2 Necessity of Agile Methodology and SCRUM

The adoption of Agile methodology and SCRUM at Brain Station 23 is a direct response to the dynamic, iterative nature of modern software development. Agile's emphasis on adaptive planning, early delivery, and continuous improvement is particularly well-suited to the organization's 15-day client update cycle and its management of frequent change requests. SCRUM, as an Agile framework, provides

a structured cadence of sprints, reviews, and retrospectives that ensures consistent progress and quality. The fact that Business Analysts fulfill the Scrum Master role at Brain Station 23 reflects the organization's recognition that requirement management and team coordination must be tightly integrated within the development process.

3.3 Necessity of Centralized Communication Platforms

The Excellent Support System (ESS) employed by Brain Station 23 fulfills the organizational necessity for a centralized communication infrastructure. In multi-stakeholder, multi-geography environments, informal communication channels are insufficient to maintain project coherence. A centralized platform ensures that all communications are documented, traceable, and accessible to authorized personnel, thereby reducing the risk of knowledge loss and miscommunication. The necessity of such a system is amplified by the organization's six-country operational footprint.

3.4 Necessity of Documentation and Quality Assurance Systems

Documentation is the institutional memory of an organization. At Brain Station 23, the necessity of comprehensive documentation — from SRS documents to sprint review records and test case logs — is rooted in the organization's ISO 9000 compliance requirements. Quality assurance systems ensure that defects are identified early, root causes are analysed, and corrective actions are systematically implemented. Without these information systems, maintaining consistent quality across a portfolio of complex, high-stakes projects would be operationally impossible.

3.5 Necessity of Project Tracking and Customer Management Tools

The 15-day client update cycle practiced by Brain Station 23 necessitates a robust project tracking and customer relationship management (CRM) system. These tools enable the organization to monitor project milestones, track deliverable completion, and generate structured progress reports for clients. The necessity of such systems is particularly acute in engagements with demanding institutional clients such as PayPal or MetLife, where accountability, transparency, and timeliness are contractual obligations.

3.6 Necessity of Information Security Management Systems (ISMS)

The ISO 27000 certification held by Brain Station 23 institutionalizes the necessity of an ISMS. In handling sensitive client data across financial, telecommunications, and retail sectors, the organization is legally and ethically obligated to deploy systems that protect data integrity, confidentiality, and availability. An ISMS encompasses policies, procedures, and technical controls that collectively constitute the organization's information security posture — a posture that is both a regulatory requirement and a competitive differentiator in the global software market.

4. Economic Value

Beyond addressing operational problems, the information systems deployed by Brain Station 23 generate substantial and measurable economic value. This section analyses the business benefits, cost implications, and long-term growth potential arising from the organization's investment in information systems.

4.1 Cost Reduction Through Process Automation

Information systems automate repetitive, labor-intensive tasks — such as status reporting, defect logging, and change request tracking — thereby reducing the human effort and associated cost required to maintain project operations. At Brain Station 23, automated project dashboards and sprint management tools eliminate the need for manual reporting, freeing business analysts and project managers to focus on higher-value strategic activities. The resultant reduction in administrative overhead translates directly into cost savings at the organizational level.

4.2 Improved Productivity and Delivery Efficiency

By providing development teams with real-time access to project requirements, design specifications, and task assignments through centralized information systems, Brain Station 23 minimizes idle time and reduces the frequency of rework caused by information gaps. The SCRUM-based sprint framework, supported by digital backlog management tools, enables the organization to deliver software increments predictably and efficiently. This improved delivery efficiency is a key differentiator in competitive client acquisition and retention.

4.3 Better Resource Utilization

Resource allocation information systems enable Brain Station 23 to match the right skills to the right projects at the right time, minimizing both underutilization and overallocation of talent. With a workforce exceeding 1,000 employees distributed across six countries, the economic value of optimized resource allocation is substantial. Reducing bench time and avoiding talent bottlenecks directly improves the organization's capacity to onboard new projects and serve existing clients more effectively.

4.4 Reduced Project Risk and Financial Exposure

Information systems significantly mitigate project risk by enabling early detection of deviations from plan, systematic management of change requests, and proactive identification of quality issues. At Brain Station 23, the formal change request queue and the 15-day review cycle function as risk management mechanisms that prevent small issues from escalating into costly project failures. The economic value of risk reduction is particularly significant in large-scale engagements where project overruns can result in substantial financial penalties and reputational damage.

4.5 Increased Customer Satisfaction and Market Competitiveness

Customer satisfaction is the ultimate economic metric for a service organization. Brain Station 23's systematic use of information systems — from SRS documentation to sprint reviews and customer portals — ensures that client expectations are consistently met and that communication is transparent and timely. High levels of customer satisfaction translate into client retention, referrals, and contract renewals, all of which contribute directly to revenue growth. The organization's impressive client portfolio, which includes Fortune 500 companies, is itself evidence of the economic value generated by its information system investments.

4.6 Long-Term Organizational Growth and Scalability

The information systems deployed by Brain Station 23 are not merely solutions to current problems; they are strategic assets that enable long-term organizational growth. By standardizing processes through ISO certifications, achieving CMMI Level 3 maturity, and leveraging AWS cloud infrastructure, the organization has built a scalable operational foundation that can accommodate growth in workforce, client base, and geographic reach without proportional increases in operational complexity or cost. This scalability is a direct consequence of deliberate, value-driven information system investment.

5. Conclusion

The industrial visit to Brain Station 23 provided invaluable firsthand insight into the practical application of Information System Design concepts within a large-scale, globally operating software organization. The visit confirmed that the Recognition of Need — the systematic identification of problems, inefficiencies, and organizational requirements that information systems must address — is not merely an academic construct but a foundational organizational practice at Brain Station 23.

The problems identified during the visit, ranging from requirement management and global team coordination to information security and quality assurance, collectively illustrate the multi-dimensional complexity of modern software organizations. Each problem was found to have a corresponding information system solution that Brain Station 23 has either implemented or is actively managing. The organization's use of SRS documents, Agile-SCRUM methodology, ESS platforms, ISO-compliant quality frameworks, and AWS cloud infrastructure collectively represent a mature, integrated response to the diverse informational needs of a large software enterprise.

From an academic perspective, the visit reinforced several key lessons. First, information systems are not isolated technical tools but holistic solutions to organizational and business problems. Second, the economic value of information systems extends far beyond cost reduction; it encompasses risk mitigation, productivity enhancement, customer satisfaction, and strategic competitive positioning. Third, the formalization of needs — through documentation, certification, and structured methodologies — is what distinguishes high-performing software organizations from those that operate reactively.

Perhaps most significantly, the visit demonstrated that the Recognition of Need must precede all other phases of information system development. An organization that cannot clearly articulate why it needs a system, what problems it must solve, and what value it must deliver is unlikely to develop a system that meets its objectives. Brain Station 23's two-decade track record of successful software delivery to global clients is, in large part, a testament to its institutional commitment to this foundational principle.

In conclusion, the industrial visit to Brain Station 23 was an intellectually enriching experience that effectively grounded the theoretical study of Information System Design in the realities of professional software engineering practice. The lessons learned will inform the students' future academic work and professional careers in the field of information systems and software development.